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A Solution to Co-occurrence Bias in Pedestrian Attribute Recognition: Theory, Algorithms, and Improvements

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Abstract
For the pedestrian attributes recognition, we demonstrate that deep models can memorize the pattern of attributes co-occurrences inherent to dataset, whether through explicit or implicit means. However, since the attributes interdependency is highly variable and unpredictable across different scenarios, the modeled attributes co-occurrences de facto serve as a data selection bias that hardly generalizes onto out-of-distribution samples. To address this thorny issue, we formulate a novel concept of attributes-disentangled feature learning, by which the mutual information among features of different attributes is minimized, ensuring the recognition of an attribute independent to the presence of others. Stemming from it, practical approaches are developed to effectively decouple attributes by suppressing the shared feature factors among attributes-specific features. As compelling merits, our method is exercised with minimal test-time computation, and is also highly extendable. With slight modifications on it, further improvements regarding better exploration of the feature space, softening the issue of imbalanced attributes distribution in dataset and flexibility in term of preserving certain causal attributes interdependencies can be achieved. Comprehensive experiments on various realistic datasets, such as PA100k, PETAzs and RAPzs, validate the efficacy and a spectrum of superiorities of our method.

Keywords Pedestrian attributes recognition · Co-occurrence bias · Attributes disentanglement · Mutual information minimization

1 Introduction

Pedestrian attribute recognition (PAR) (Wang et al., 2022; Li et al., 2023) gives structured description of an individual's main-body image, by identifying its soft-biometrics of local or high-level attributes. Due to its large potential for applications in intelligent video surveillance (Yang Yang & Li, 2021; Ran et al., 2007), and intrinsic connection to other pedestrian analysis tasks, such as person re-identification (Yutian et al., 2019; Zhang et al., 2021) and person retrieval (Li et al., 2019), great efforts have been devoted to this field.

In PAR, a thread of literature (Wang et al., 2016; Li et al., 2022) regards attributes co-occurrence, which is embedded in dataset, as a contextual constraint complementary to visual attributes recognition. Under such assumption, graph-based models are commonly adopted to explicitly model the joint label probability, by which attributes are inferred. However, the inherent variability and unpredictability of attributes

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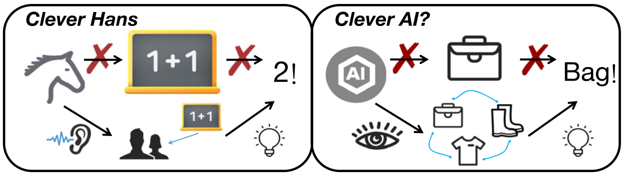

Fig. 1 Illustration of the main motivation of this paper. By recognizing certain co-occurred behaviors of bystanders, the pony Hans does calculations accurately, analogous to that deep models rely on attributes co-occurrences to precisely recognize the attributes. However, what if there are no people nearby Hans, or the pattern of attributes interdependencies shifts? We care about the problem that, *does Hans really calculate, or does current AI system really learn to recognize attributes, if other co-occurred factors are not disregarded*

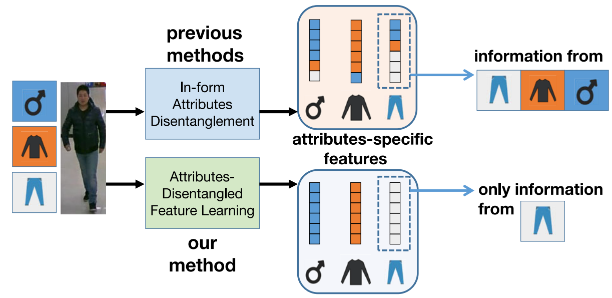

Fig. 2 Concept of attributes-disentangled learning. Instead of decomposing attribute features merely in forms, our method enables semantics extracted from one attribute not being applied to infer other attributes

co-occurrences raise concern regarding the robustness of these methodologies in scenarios distributionally far from the dataset. As such, these methods can encounter difficulties in generalizing onto scenes or identities with attributes interdependencies gap.

Equipped with above perspectives, this paper approaches PAR in an unconventional manner. We regard such memorizable attributes co-occurrence as a form of data selection bias, and aim to infer attributes while disregarding any potential relations among them. This mechanism actually mimics human behavior, as we do not infer an individual's hat by referencing its gender. Instead, we rely solely on visual cues of the hat for nimble and interdependencies-agnostic identification. Similarly, we believe that models should be intelligent at not only telling us *what* (recognition results), but also knowing sense-making *why* (there are no invariable and universal attributes interdependencies), in case of being a *Clever Hans* (Kauffmann et al., 2020) that calculates by recognizing onlookers expressions instead of math, illustrated in Fig. 1.

To embody this philosophy, we formalizes a theoretically well-defined concept of attributes-disentangled feature learning, that the specific feature learned for predicting one attribute should not utilize any information pertaining to other attributes. In other words, the mutual information between attributes specific features should be minimized. By doing so, we can achieve interdependencies-free inference as no common semantics are shared for identifying different attributes (refer to Fig. 2), serving as the driving principle of this paper.

Practically, we begin by introducing a simple yet strong baseline method, termed as Donsker-Varadhan representation (Donsker & Varadhan, 1975) based Mutual Information Minimizer (DVMIM), which directly minimizes the underlined mutual information by representing it as the Kullback–Leibler (KL) divergence between the joint distribution and the product of the marginals (Belghazi et al., 2018). Despite the plainness of DVMIM, it still excels prior arts with margins, validating effectiveness of the attributes-disentangled learning. However, DVMIM trains the model in a generative adversarial manner with ever-shifted target distribution, suffering in poor convergence stability and training efficiency.

To facilitate optimization-amenable learning of disentangled features, we sequentially propose an enhanced method named Posterior-Invariant Learning (PIL). On each of the attribute-specific feature, PIL imposes a random and independent in-distribution transform on it. By requiring that any potential variation of a certain attribute's specific feature gives no knowledge predictive to other attributes, PIL ensures that no feature factors are shared across attributes.

The superiorities of PIL are supported from both mathematical insights and experimental evidence. Specifically, PIL enables the generation of attributes-disentangled features, effectively mitigating the inherent bias arising from attributes co-occurrence. Furthermore, PIL exhibits high flexibility and extensibility. With minor modifications, it can be refined to accomplish the followings: Mitigating the accuracy degradation caused by imbalanced label distribution in dataset; Preserving the information exchange and propagation among attributes with causal relations while still disregarding modeling the rest co-occurrences bias; Enabling a novel technique called Norm-Direction Separated Interpolation (NDSI) to enhance the exploration of the distribution of attributes features. Importantly, unlike existing methods that rely on large framework, PIL is highly lightweight and efficient, catering to practical applications with limited resource. The following aspects distinguish this paper from previous works:

- We provide solid discussion and experimental evidence for the novel perspective on interdependency bias of learned attributes. It bridges a gap in existing researches and can serve as a new mindset for future work.
- Two information-theoretic approaches are developed to learn the desirable attributes-disentangled features. Serval improvements on the proposed method to further boost its performance are also introduced.
- We present analytical experiments on a suite of realistic PAR datasets to demonstrate the state-of-the-art efficacy of our methods and a range of their superiorities.

This paper is an extension of our previous work (the original conference paper is placed in the supplementary). We have made several significant improvements, majorly including a new method called DVMIM to provide a ground baseline for attributes-disentangled learning and a quantitative measurement of the inference interdependencies among attributes (Sect. 4.2), new modifications on the proposed method PIL to further boost its performance (Sect. 4.4), comprehensive discussion and theoretical analyses on disentangled attributes feature learning (Sects. 5.3 and 5.4) and more experimental justification (Sects. 5.2, 5.3 and 5.4).

2 Related Work

2.1 Pedestrian Attribute Recognition

Hitherto, researches on PAR have basically followed the routine of discriminative multi-label classification and studies have generally delved into improving attributes localization to mitigate accuracy loss from inferring on irrelevant areas (Liu et al., 2018; Jia et al., 2022; Shen et al., 2024). To utilize the prior knowledge of the body topology for better attribute localization, (Fabbri et al., 2017; Li et al., 2017) do not consider the person as a whole. Instead, they split the body image vertically into three parts with a manually defined strategy. Each portion is then fed into an individual network to extract features. Similarly, (Liu et al., 2018) guides PAR with pose key points. To further enhance attributes localization, (Liu et al., 2017) proposes the HydraPlus-Net to locate attributes of various scales by multi-directional attention modules. Rather than learning to generate the attention map, (Liu et al., 2018) produces attributes-specific features with the class activation map (CAM) (Zhou et al., 2016), which is initially designed to develop a generic localizable deep representation. However, same technique is applied in (Jia et al., 2021b) to demonstrate that, even without explicitly modeling the attributes-specific area, deep model can still locate attributes with precision, highlighting that the fundamental task in PAR should be on better feature learning. Other works in PAR guide model with external information or supervision, like body key points (Liu et al., 2018), pedestrian orientation (Lu et al., 2023) and pedestrian video clip (Chen et al., 2019; Ji et al., 2020), etc., under explicit assumptions considering certain restrictions from body topology or temporal context. Differently, in this work, we do not enhance PAR by other visual tasks, and nor do we involve extra information of data sources (Wang et al., 2024; Zhu et al., 2023).

2.2 Attributes Disentanglement

Previous methods in PAR generally consider the joint attributes probability in dataset as known a-priori knowledge (Wang et al., 2017; Zhu et al., 2017; Fan et al., 2020), and aim to explicitly infer attributes with it. Advancing in an opposite direction, this work is the first attempt in PAR to recognize attributes by totally discarding any potential pattern of co-occurrences among them. This concept of attributes disentanglement is inspired by the development of zero-shot learning (Lampert et al., 2009; Yongqin et al., 2017) and attributes-based generation (Yan et al., 2016; Lample et al., 2017). In zero-shot learning literature, attributes disentanglement is proposed to recognize unseen categories by only given their class attributes (Yue et al., 2021), while attributes-based generation and conditional generation leverage attributes disentanglement mainly for interpretable or manipulatable representations (Zhu et al., 2021). These methods approach disentanglement from different sides. The first type is to model the informativeness in latent embeddings, such as InfoGAN (Chen et al., 2016) and IB-GAN (Jeon et al., 2021), which optimizes upper bounds of the informativeness on a subset of latent variables and the produced images. Another class of methods models the statistical independence in the feature embeddings with VAE framework (Higgins et al., 2017; Jeon et al., 2021), which minimizes the correlation in learned features by factorizing the aggregated posterior. Contrastive learning (Yue et al., 2021) is also adopted into attributes disentanglement to comply with the definition of causal disentanglement (Do & Tran, 2019). Albeit effective, these methods are majorly designed for unsupervised learning or image generation, and thus can not be directly transferred into the topic of PAR. Also, as far as we know, disentangling attributes by minimizing the mutual information is not covered by prior arts. Therefore, attribute feature disentanglement via an objective of mutual information minimization is both novel in PAR and disentanglement literature.

3 On the Attributes Co-occurrence Bias

Intrinsically, while there may exist a pattern of interdependencies among attributes, this phenomenon is rooted in *conditioned statistical relevance* rather than *causality*. It implies that the co-occurrences statistics drawn from limited scenarios, such as datasets that solely involve indoor or outdoor scenes, a constant season or weather condition (Li et al., 2016; Liu et al., 2017), can serve as immense data selection bias that varies drastically from scene to scene, time to time and individual to individual.

This claim is supported by the observations depicted in Fig. 3, which illustrate the variability and unpredictability in attribute correlations between different datasets such as PA100k (Liu et al., 2017) and PETA (Deng et al., 2014), as well as different groups of identities from the same scenery in PETA. For example, in PA100k, there is a tendency for ShortSleeve and Trouser attributes to not co-occur, likely due to the

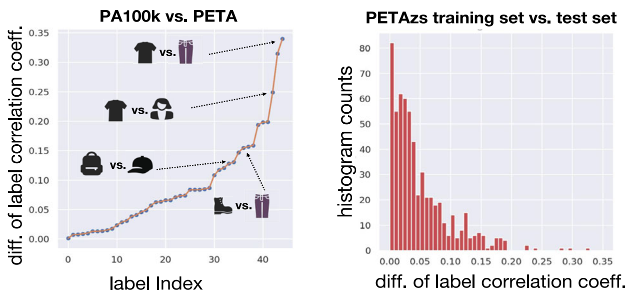

Fig. 3 Attributes correlation exhibits great variability. Left: We compute the labels Pearson correlation coefficients between 10 common attributes for PA100k and PETA, respectively, and plot the corresponding absolute differences between them. Values are sorted by an increasing order. Right: For the PETAzs dataset (no overlapped identities of training set in test set), distribution of the absolute differences in labels correlation coefficients of training and test set

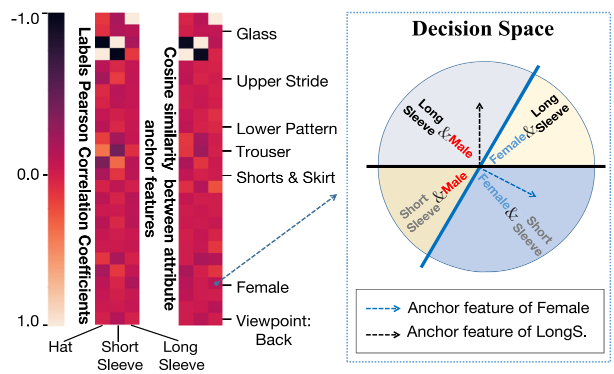

Fig. 4 Even a baseline model can exactly capture the biased attributes correlation in dataset and infer attributes with it. We demonstrate this by showing that for ShortSleeve, LongSleeve and Hat *vs.* other attributes in PA100K, the Pearson correlation coefficients of labels, and the cosine similarities of learned anchor features (weights of the last fully-connected layer in a Resnet-50) exhibit a similar pattern. Actually, each anchor feature in the classifier directs a hyper-plain serving as the decision boundary of one attribute. Thus, the larger the cosine similarity between two attributes anchor features, the larger the intersection between their respective decision space of half space. In others words, PAR model can infer with the co-occurrences pattern in dataset, by simply modulating the overlap of attributes decision spaces

majority of images being captured during the summer season. However, for PETA, ShortSleeve and Trouser attributes co-occur with a higher frequency, resulting in a notable discrepancy of attributes interdependency between these two datasets. Furthermore, even between two random splits of pedestrians from PETAzs (Jia et al., 2021b), a considerable portion of attributes correlations exhibit distinct characteristics. Aggravating this problem, such biased interdependency can be memorized by network even in the absence of explicit modeling of it, as evidenced in Fig. 4, which puts all prior methods at the empirical risk of co-occurrences bias.

To circumvent this problem, collecting a dataset of pedestrian images that captures global facets of the non-static myriad population distribution, however, is infeasible. As a result, for a generalizable PAR, learning disentangled and discriminative feature for each attribute becomes indispensable. Besides, AI models should reflect as less information as possible of the adopted datasets to embrace the Holy Grail of unbiased intelligence, especially task like PAR that entails heavy modeling of the structured privacy information, should be researched under the consideration of bias minimization to avoid information leakage in any potential forms, of any specific group of identities. This paper contributes in this perspective, by showing that the interest of high performance PAR can be actually aligned to that of the privacy protection of individual attributes co-occurrence, which is vital for the acceptability of society to PAR and can potentially shed light onto the long run of this community.

4 Method

We begin this section by formalizing the definition of attributes-disentangled features. Towards such features, sequentially we propose two information-theoretic regularizers, namely Donsker-Varadhan representation based Mutual Information Minimizer (DVMIM) and Posterior-Invariant Learning (PIL). Lastly, we present serval simple modifications on the PIL to get further performance boost.

4.1 Disentangled Attributes Features

Formally, X is a distribution characterized by pedestrian surveillance images. Images $\{\boldsymbol{x}_i\}_{i=1}^N$ are sampled from X, with their labels $\{\boldsymbol{a}_i\}_{i=1}^N$ of predefined attributes, to form the training set $D := \{(\boldsymbol{x}_i, \boldsymbol{a}_i)\}_{i=1}^N$, where $\boldsymbol{a}_i \in \{0, 1\}^C$ and C is the number of attributes. Each element in $\boldsymbol{a}_i$ indicates the existence of an attribute in $\boldsymbol{x}_i$. For a PAR model, if the feature extracted from the backbone is $\boldsymbol{f} \in \mathbb{R}^K$, we decompose it as $\mathcal{H}_\psi(\boldsymbol{f}) = \{\boldsymbol{f}^1, \boldsymbol{f}^2, \ldots, \boldsymbol{f}^C\}$, where $\boldsymbol{f}^s \in \mathbb{R}^K$ is the feature specific to predict the posterior probability of an attribute $y^s := P(\boldsymbol{a}^s = 1|\boldsymbol{x})$, $s = 1, 2, \ldots, C$, and $\mathcal{H}_\psi(\cdot)$ is a neural function parameterized by ψ.

In this work, we aim to disentangle attributes by the information theory. Mutual information $\mathcal{I}(a; b)$ is a Shannon entropy-based measure of common knowledge between two random variables a and b. In contrast to correlation, it captures non-linear statistical relation, and thus measures the true dependencies between variables (Kinney & Atwal, 2014). Therefore, to ensure that an attribute specific feature exploits no clues indicating the presence of other attributes, the objective of this work is to make $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s) = 0, s = 1, 2, \ldots, C$ satisfied, where $\boldsymbol{y}^{/s} := (y^1, y^2, \ldots, y^{s-1}, y^{s+1}, \ldots, y^C)$. For this, we propose two practical approaches in the following.

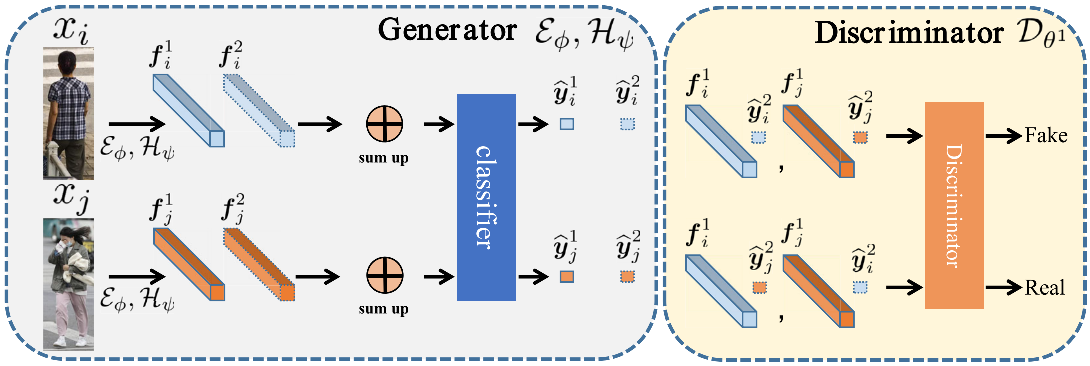

Fig. 5 Illustration for the main idea of DVMIM. The generator produces attributes-specific features that deliver no knowledge about the existence of other attributes, by fooling a discriminator that learns to discern between feature-posterior pairs sampled from the joint and product of marginals

4.2 Donsker-Varadhan Representation Based Mutual Information Minimizer

One intuition is that the larger the discrepancy between the joint and the product of the marginals of two random variables, the larger the dependencies between them (Nguyen, 2010; Ruderman et al., 2012; Belghazi et al., 2018). Therefore, mutual information $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s)$ can be represented as the KL divergence between the joint probability of $\boldsymbol{y}^{/s}\&\boldsymbol{f}^s$, and the product of their marginals

$$\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s) = D_{KL}(P(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)||P(\boldsymbol{y}^{/s}) \times P(\boldsymbol{f}^s)). \quad (1)$$

If the KL divergence in Eq. (1) vanishes to 0, s^{th} attribute's specific feature $\boldsymbol{f}^s$ acquires no factors informative about other attributes. In other words, the feature factors to infer the s^{th} attribute are individualized from those of other attributes. To minimize Eq. (1), we resort to the Donsker-Varadhan representation (Donsker & Varadhan, 1975). Specifically,

Theorem 1 (Donsker-Varadhan representation) *The KL divergence admits the dual variational representation of:*

$$D_{KL}(P||Q) = \sup_{T:\Omega \mapsto \mathbb{R}} \mathbb{E}_P[T] - log(\mathbb{E}_Q[e^T]), \quad (2)$$

where the supremum is taken over all functions T such that the two expectations are finite.

With the Donsker-Varadhan representation, the disentangled attribute-specific feature $\hat{\boldsymbol{f}}^s$ can be generated by

$$\begin{aligned}\hat{\boldsymbol{f}}^s &= \arg\min_{\boldsymbol{f}^s} D_{KL}(P(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)||P(\boldsymbol{y}^{/s}) \times P(\boldsymbol{f}^s)) \\ &\approx \arg\min_{\boldsymbol{f}^s} \max_{\theta} \mathbb{E}_{(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)\sim P(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)}[\mathcal{D}_\theta(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)] \\ &\quad - log(\mathbb{E}_{(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)\sim P(\boldsymbol{y}^{/s})\times P(\boldsymbol{f}^s)}[e^{\mathcal{D}_\theta(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)}]),\end{aligned} \quad (3)$$

where $\mathcal{D}_\theta(\cdot, \cdot)$ is a θ parameterized discriminator. Practically, we implement Eq. (3) for PAR by minimizing the loss $\mathcal{L}_{DV}$,

$$\mathcal{L}_{DV} = \mathcal{L}_{cls} + \mathcal{L}_{dis} + \lambda \cdot \mathcal{L}_{gen}. \quad (4)$$

Here, $\mathcal{L}_{cls}$ is the cross entropy for multi-label classification

$$\begin{aligned}&\mathcal{L}_{cls} = \min_{\psi,\phi,\omega} -\sum_{i=1}^{N}\sum_{s=1}^{C} \boldsymbol{a}_i^s \log \widehat{\boldsymbol{y}}_i^s, \\ &s.t.\ \widehat{\boldsymbol{y}}_i = \mathcal{R}_\omega\left(\sum_{k=1}^{C} \boldsymbol{f}_i^k\right) \quad and \quad \{\boldsymbol{f}_i^k\}_{k=1}^C = \mathcal{H}_\psi(\mathcal{E}_\phi(\boldsymbol{x}_i)).\end{aligned} \quad (5)$$

$\mathcal{E}_\phi(\cdot)$ is a ϕ parameterized backbone of feature extractor, $\mathcal{R}_\omega(\cdot)$ is a fully-connected (FC) layer, followed by the sigmoid operation to estimate the attributes posterior $\widehat{\boldsymbol{y}}_i$, $H_\psi(\cdot)$ is a FC layer to decompose the extracted feature into attributes-specific features. $\mathcal{L}_{dis}$ in Eq. (4) is a looser estimate of the underlined mutual information, which leverages the universal expressive power of neural network, and $\mathcal{L}_{gen}$ produces attributes-disentangled features by adversarially minimizing $-\mathcal{L}_{dis}$. Concretely,

$$\begin{aligned}\mathcal{L}_{dis} = &\min_{\{\theta^k\}_{k=1}^C} \sum_{i=1}^{N}\sum_{s=1}^{C} -\mathcal{D}_{\theta^s}(\odot(\boldsymbol{f}_i^s, \widehat{\boldsymbol{y}}_i^{/s})) \\ &+ \log \sum_{i=1}^{N}\sum_{s=1}^{C} e^{\mathcal{D}_{\theta^s}(\odot(\boldsymbol{f}_i^s, \widehat{\boldsymbol{y}}_{\neq i}^{/s}))},\end{aligned} \quad (6)$$

and $\mathcal{L}_{gen} = \min_{\psi,\phi} -\mathcal{L}_{dis}$, where $\odot$ is vector concatenation. We employ total C discriminators, one for each attribute. The expectations taken over the joint in Eq. (3) are estimated by sampling $\boldsymbol{y}^{/s}$ and $\boldsymbol{f}^s$ from a same image. While for $(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)$ drawn from the marginals, we approximate it by shuffling the joint distribution along the batch axis, i.e., $\boldsymbol{f}^s$ and $\boldsymbol{y}^{/s}$ are respectively generated from two random images within a batch, graphed in Fig. 5.

Also, the value of $-\mathcal{L}_{dis}$ can act as a measurement of $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s)$, based on which we can quantitatively analyse the inference dependence on a given attribute's feature. It is not entirely new as (Belghazi et al., 2018) has pioneered a similar form. However, to the best of our knowledge, this is the first attempt to utilize the Donsker-Varadhan representation of mutual information in a form of min-max game to learn the disentangled features, and we believe that DVMIM can be a validated baseline method for attributes-disentangled learning in PAR, considering its clarity in term of the underlined mutual information minimization.

4.3 Posterior-Invariant Learning

Here, we propose an improved method that minimizes the mutual information without adversarial generative training. At first, we declare the posterior-invariant learning (PIL).

Theorem 2 (Posterior-invariant learning) *Disentangled attributes features can be enabled once $P(\boldsymbol{y}^{/s}|\sum_{l=1}^{C}\boldsymbol{f}^l)$ is invariant to $\boldsymbol{f}^s$, or equivalently, if $P(\boldsymbol{y}^s|\sum_{l=1}^{C}\boldsymbol{f}^l)$ varies with and only with its own specific feature $\boldsymbol{f}^s, \forall s = 1, 2, \ldots, C$.*

Proof $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s) = 0$ can be succinctly reformulated as

$$\begin{aligned}\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s) &= \mathcal{H}(\boldsymbol{y}^{/s}) - \mathcal{H}(\boldsymbol{y}^{/s}|\boldsymbol{f}^s)\\ &= \mathbb{E}_{\boldsymbol{f}^s\sim\mathcal{F}^s}\left[\int P(\boldsymbol{y}^{/s}|\boldsymbol{f}^s)\log P(\boldsymbol{y}^{/s}|\boldsymbol{f}^s)\,d\boldsymbol{y}^{/s}\right]\\ &\quad-\int P(\boldsymbol{y}^{/s})\log P(\boldsymbol{y}^{/s})\,d\boldsymbol{y}^{/s} = 0,\end{aligned} \tag{7}$$

where $\mathcal{F}^s$ is the distribution of $\boldsymbol{f}^s$. Notably, if $P(\boldsymbol{y}^{/s}|\boldsymbol{f}^s) = P(\boldsymbol{y}^{/s})$, Eq. (7) naturally holds as $\int P(\boldsymbol{y}^{/s})\log P(\boldsymbol{y}^{/s})\,d\boldsymbol{y}^{/s}$ is invariant to $\boldsymbol{f}^s$. Thus, if $P(\boldsymbol{y}^{/s}|\boldsymbol{x}) = P(\boldsymbol{y}^{/s}|\sum_{l=1}^{C}\boldsymbol{f}^l)$ is independent of $\boldsymbol{f}^s$, we have $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s) = 0$. Towards this aim, in practice, for a given input $\boldsymbol{x}_i$, we decompose its extracted one-dimensional embedding $\boldsymbol{f}_i = \mathcal{E}_\phi(\boldsymbol{x}_i) \in \mathbb{R}^K$ into attributes-specific features with a FC layer: $\{\boldsymbol{f}_i^s\}_{s=1}^C = \mathcal{H}_\psi(\boldsymbol{f}_i)$. Sequentially, we construct C random transforms $\mathcal{G}_i^s(\cdot) : \mathcal{F}^s \mapsto \mathcal{F}^s, s = 1, 2, \ldots, C$ to independently map every $\boldsymbol{f}_i^s$ into an arbitrary feature point $\tilde{\boldsymbol{f}}_i^s = \mathcal{G}_i^s(\boldsymbol{f}_i^s)$ within the same domain of $\mathcal{F}^s$. We require $P(\boldsymbol{y}_i^s| - \tilde{\boldsymbol{f}}_i^s + \boldsymbol{f}_i^s + \sum_{l=1}^{C}\tilde{\boldsymbol{f}}_i^l) = P(\boldsymbol{y}_i^s|\sum_{l=1}^{C}\boldsymbol{f}_i^l)$ to render $\boldsymbol{y}^s$ tolerant to any in-distribution variation of $\boldsymbol{f}^k, \forall k \neq s$, i.e., no factors predictive to $\boldsymbol{y}^s$ are conveyed by $\boldsymbol{f}^k$, meeting the property of the attributes-disentangled features. Specifically, we leverage an information-theoretic regularizer of

$$\begin{aligned}&\mathcal{L}_{PIL} = \min_{\psi,\phi,\omega} -\sum_{i=1}^{N}\sum_{s=1}^{C}(\alpha^s\boldsymbol{a}_i^s + (1-\alpha^s)\boldsymbol{a}_{r(i,s)}^s)\log\widehat{\tilde{\boldsymbol{y}}}_i^s.\\ &s.t.\ \widehat{\tilde{\boldsymbol{y}}}_i = \mathcal{R}_\omega(\sum_{s=1}^{C}\tilde{\boldsymbol{f}}_i^s),\quad \tilde{\boldsymbol{f}}_i^s = \alpha^s\boldsymbol{f}_i^s + (1-\alpha^s)\boldsymbol{f}_{r(i,s)}^s,\\ &\qquad\{\boldsymbol{f}_i^s\}_{s=1}^C = \mathcal{H}_\psi(\mathcal{E}_\phi(\boldsymbol{x}_i))\ and\ \{\alpha^s\}_{s=1}^C \sim \mathcal{U}(0,1),\end{aligned} \tag{8}$$

where $\widehat{\tilde{\boldsymbol{y}}}_i$ is an estimate of the posterior of $\sum_{s=1}^{C}\tilde{\boldsymbol{f}}_i^s$ given by a discriminative classifier $\mathcal{R}_\omega(\cdot)$. For the specific form of $\mathcal{G}_i^s(\cdot)$, we opt for simplicity by employing the convex linear combination to generate the in-distribution transformed feature $\tilde{\boldsymbol{f}}_i^s = \mathcal{G}_i^s(\boldsymbol{f}_i^s) = \alpha^s\boldsymbol{f}_i^s + (1-\alpha^s)\boldsymbol{f}_{r(i,s)}^s$, where $\boldsymbol{f}_{r(i,s)}^s$ is the attribute-specific feature extracted from a randomly drawn training data $\boldsymbol{x}_{r(i,s)}$, α^s is an interpolation weight sampled from uniform distribution $\mathcal{U}(0, 1)$, and importantly, $\alpha^k \neq \alpha^s, \forall k \neq s$. Intuitively, different training stages (initial vs. converged) yield not identical $\mathcal{F}^s$ due to the update of parameters of ϕ and ψ. Therefore, $r(i, s)$ is applied as a function to map different i&s to a different batch sample index. In other words, features are fused within a batch. Notably, with plain vector combination, points off the training distribution can be produced. To secure the closeness of the transform, say, the interpolated feature $\tilde{\boldsymbol{f}}_i^s$ still lies within $\mathcal{F}^s$, we guide the model to generate features $\boldsymbol{f}^s$ interpolatable to produce meaningful and informative in-distribution semantics. To be specific, the well-known method MixUp (Zhang et al., 2017) shows that the random interpolation between images in pixel space can be aligned with exactly the same linear combination of their ground truth in the label space. Here, in Eq. (8) we adopt an analogous idea onto the attributes-specific features by aligning the interpolated feature $\alpha^s\boldsymbol{f}_i^s + (1-\alpha^s)\boldsymbol{f}_{r(i,s)}^s$ with an in-distribution supervision signal, i.e., its correspondingly interpolated label $\alpha^s\boldsymbol{a}_i^s + (1-\alpha^s)\boldsymbol{a}_{r(i,s)}^s$, tasking the fused feature pertinent to its corresponding attribute and thus reside in $\mathcal{F}^s$.

Equation (8) actually differs from MixUp in two aspects. First, MixUp conducts a single interpolation on the whole image, merely for the sake of data augmentation. Whereas in Eq. (8), on each attribute specific feature $\boldsymbol{f}_i^s$, we impose feature interpolation, with independent α^s and $\boldsymbol{f}_{r(i,s)}^s$, as a transform differentiated from that of others. By enforcing the variation of a certain attribute posterior exclusively aligned to the variation of its specific feature, we make other attributes features acquire no factors indicative to the given attribute posterior, realizing that $P(\boldsymbol{y}^s|\sum_{l=1}^{C}\boldsymbol{f}^l)$ varies with and only with its own specific feature $\boldsymbol{f}^s$. In other words, model trained with Eq. (8) learns to produce feature factors unshared among attributes. Thus, although features are re-summed before being fed into the classifier, each of the semantic factors is attribute-unique. It truly matters than decomposing merely

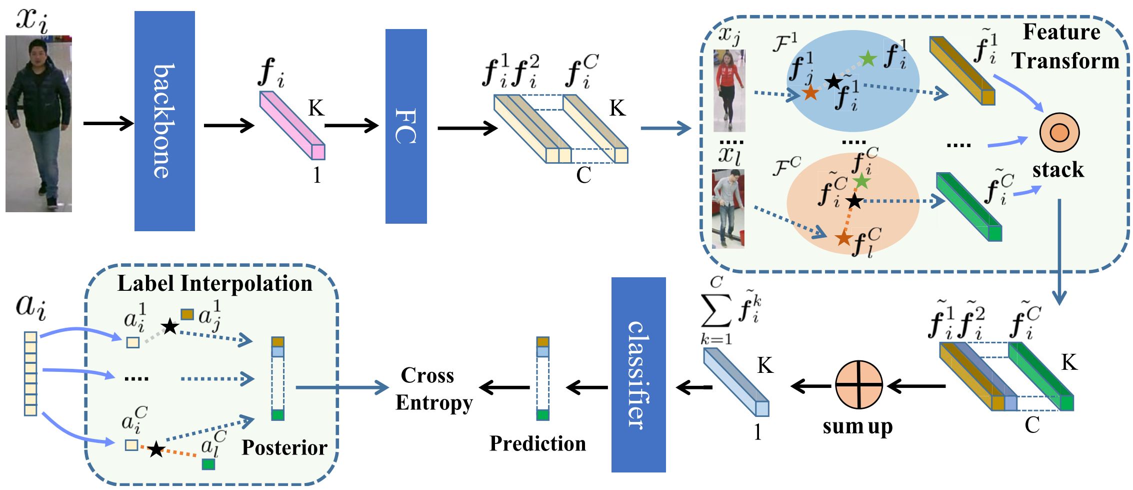

Fig. 6 The overall training framework of PIL. $\boldsymbol{x}_i$, $\boldsymbol{x}_j$ and $\boldsymbol{x}_l$ are sampled within a batch

in form and applying a separate classifier on each of the $\boldsymbol{f}^s$, like previous work (Jia et al., 2022), which can still embed shared information among attributes features. Hence, as an additional merit of Eq. (8), we can only use one classifier with the re-summed features, $\sum_{s=1}^{C} \tilde{\boldsymbol{f}}_i^s$ for training and $\sum_{s=1}^{C} \boldsymbol{f}_i^s$ for testing, as input to save model parameters.

Second, in contrast to images interpolation, Eq. (8) operates in the space of decoupled features. While one may argue that this approach can enhance performance by acting as a form of data augmentation akin to MixUp, we acknowledge this concern. However, our experimental results indicate that it is not the case for achieving state-of-the-art performance, as MixUp fails to deliver any accuracy beyond the baseline. Moreover, we demonstrate in the experiment section that the DVMIM can produce results similar to PIL, which does not optimize over any interpolated features.

Furthermore, seamlessly integrated into the framework graphed in Fig. 6, PIL enjoys another virtue whereby memorization of the biased attributes interdependencies is eliminated from classifier training. This is achieved by learning over the ground truth of $\alpha^s \boldsymbol{a}_i^s + (1 - \alpha^s)\boldsymbol{a}_{r(i,s)}^s$ that is independently generated for each attribute, encompassing all possible pattern of attribute co-occurrence. As a result, the classifier cannot trace the limited attributes correlations in dataset, enabling it better generalize to scenarios with out-of-distribution pattern of attributes co-occurrences. □

4.4 Further Improvements on Posterior-Invariant Learning

We demonstrate the extensibility of PIL by showing that it can be further improved by several simple modifications. Importantly, all the following techniques can be implemented in a holistic manner with the training of PIL.

Attributes-Balanced Re-sampling. PAR datasets usually exhibit severely imbalanced labels distributions, where some minority attributes are under-represented while some majority attributes can present in most of the images. Actually, it is non-trivial to over sample the images of a certain minority attribute to balance its labels distribution, while not replicating the attributes co-occurrence pattern embedded in them, on which serious overfitting of attributes interdependencies can be delivered.

To cope with this issue, based on PIL, we propose the method of attributes-balanced re-sampling. Concretely, we first build C pairs of empty memory banks $\{(O_0^s, O_1^s)\}_{s=1}^{C}$, where O_0^s and O_1^s are memory banks to preserve the disentangled feature $\boldsymbol{f}^s$ of label 0 and 1, respectively. For PIL, when the training of $\mathcal{E}_\phi(\cdot)$ and $\mathcal{H}_\psi(\cdot)$ converges, we keep running them for additional epochs and save the produced disentangled feature $\boldsymbol{f}^s$ into the O^s that corresponds to its label. Once every memory bank is fulfilled to the maximum size, we adjust the sampling function in Eq. (8) from the within-batch random sampling $r(i, s)$ to the attributes-balanced sampling $b(i, s)$ and only fine tune the classifier, where $b(i, s)$ makes each of O_0^s and O_1^s have an equal probability to be selected, and sequentially a feature from the chosen memory bank is uniformly drawn. By doing so, the attribute information passed into the classifier is aptly balanced between labels.

Norm-Direction Separated Interpolation. In order to fully explore the domain $\mathcal{F}_i$ and generate feature points with potentially meaningful semantics in the capricious environment, we perform linear combinations on the norm and direction vector of attribute-specific features that have identical labels, respectively. Concretely,

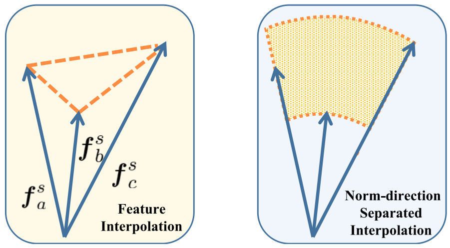

Fig. 7 Illustration of the proposed norm-direction separated interpolation (NDSI). For three features with identical labels, with vanilla feature interpolation (left) only points along the orange lines can be interpolated as possible variations. Whilst for NDSI (right), the possible variations are extended to the whole orange-shaded area in a reliable manner

$$\mathcal{G}_i^s(\boldsymbol{f}_i^s) = \begin{cases} \alpha^s \boldsymbol{f}_i^s + (1-\alpha^s)\boldsymbol{f}_{r(i,s)}^s \quad if\ \boldsymbol{a}_i^s \neq \boldsymbol{a}_{r(i,s)}^s \\ \\ (\beta^s \|\boldsymbol{f}_i^s\| + (1-\beta^s)\|\boldsymbol{f}_{r(i,s)}^s\|) \\ \cdot \left(\alpha^s \frac{\boldsymbol{f}_i^s}{\|\boldsymbol{f}_i^s\|} + (1-\alpha^s)\frac{\boldsymbol{f}_{r(i,s)}^s}{\|\boldsymbol{f}_{r(i,s)}^s\|}\right) \quad o/w, \end{cases} \tag{9}$$

where α^s and β^s are independently sampled from $\mathcal{U}(0, 1)$ for each attribute. Equation (9) is based on the reasoning that feature norm is a validated measure of data uncertainty (Shalev et al., 2018; Meng et al., 2021; Zhou, 2022), which is often affected by poor image quality (commonly observed in PAR). Therefore, we hypothesize that the style semantics of attributes such as color, size, and shape are predominantly encoded in feature direction, i.e., the normalized unit feature (Wang et al., 2018). Thus, by separately transforming norm and direction, we can generate attribute features of a particular style but have varying levels of uncertainty (or the opposite), and thereby explore more of $\mathcal{F}_s$, exemplified in Fig. 7.

Preserving Causal Interdependencies. Seemingly, PAR should infer a-priori concrete attributes interdependencies while disregarding the others. For example, the mutual exclusivity among age attributes, like Age≤18, 18<Age<60, and 60≤Age in PA100k, implies that for each input image, only one age prediction should be made. However, the indiscriminative discarding of all attributes interdependencies in Eq. (8) might give irrational predictions, like a person is predicted as both Age≤18 and Age≥ 60, impairing model's reliability regarding common sense.

By adjusting the interpolation weight α^s in Eq. (8), we can introduce export knowledge into the training of PIL and preserve the information exchange among certain attributes that are causally bound. Taking the age attributes in PA100k as an instance, we can simply set identical interpolation weights for their features and labels in Eq. (8), to save the common feature factors and not give irrational labels. By doing so, the mutual exclusivity among age attributes is retained in feature learning, whilst other attributes remain decoupled. Hence, PIL is flexible in term of the removal of attributes interdependencies. Further investigation on this topic is presented in the subsequent experiment section.

5 Experiments

5.1 Experiment Settings

Datasets: Our experiments strictly follow the benchmark protocol given in (Jia et al., 2021b). Specifically, we adopt the pedestrian attribute datasets of PETA (Deng et al., 2014), RAP (Li et al., 2016) and PA100k (Liu et al., 2017) for testing. We also evaluate our methods on two realistic datasets RAPzs and PETAzs (Jia et al., 2021b), which are respectively formed from RAP and PETA to comply with the zero-shot setting of pedestrian identities, i.e., no identities are overlapped between the training and test set. Detailed description and usage of each employed dataset can be found in (Jia et al., 2021b). The evaluation metrics in our study include the label-based mean Accuracy (mA), which calculates the average of all attribute classification accuracy on positive and negative samples, and three instance-based metrics Recall, Precision (Prec.) and F1-score (F1).

Models Architecture: To examine the compatibility and efficacy of our method under backbones of various designs, we respectively utilize the Resnet-50 (He et al., 2016) and ConvNeXt-base (Liu et al., 2022) as the feature extractor $\mathcal{E}_\phi(\cdot)$ in the benchmark experiments. $\mathcal{R}_\omega(\cdot)$ is simply a linear classifier, $\mathcal{H}_\psi(\cdot)$ consists of a single FC layer with output size of $C \times K$, followed by a nonlinearity. We opt for simplicity by implementing the discriminators $\mathcal{D}_{\theta^s}(\cdot,\cdot)$, $s = 1, 2, \ldots, C$, in DVMIM as multilayer perceptron networks.

Training Details: Across all settings, we utilize the Adam solver without Nesterov momentum. Input image is resized to the spatial resolution of 256×192. Random image crop and horizontal flip of probability 0.5 are the only data augmentations. Weight decay is applied as 5e-4, and we do not use dropout or other refinement tricks. The learning rate begins at 1e-4 (6e-5 for RAPzs) and decays by a factor of 10 in a multistep manner. The learning rate for classifier fine-tuning in attributes-balanced re-sampling is a constant of 1e-5 (6e-6 for RAPzs). Batch size and epoch number are 64 and 80, respectively. In this section, unless otherwise stated, PIL with norm-direction separated interpolation and attributes-balanced re-sampling is used to produce the results. Reasons of not considering causal interdependencies preservation as a default setting are discussed in ablation studies.

Table 1 Benchmark results on PA100k, RAP, RAPzs, PETA and PETAzs

Method	Backbone	PA100k		RAP		RAPzs		PETA		PETAzs	
		mA	F1	mA	F1	mA	F1	mA	F1	mA	F1
Baseline ('21)	Resnet-50	80.38	87.05	80.32	79.46	72.32	76.75	84.42	85.97	71.62	71.68
MsVAA ('18)	Resnet-50	80.41	86.80	78.86	79.27	72.04	75.74	83.57	85.64	71.53	71.94
MTMS ('19)	Resnet-50	–	–	82.45	65.33	–	–	86.23	85.85	–	–
VAC ('20)	Resnet-50	80.11	86.45	80.27	78.36	73.70	76.12	84.58	85.83	71.91	70.90
*JLAC ('20)	Resnet-50	82.31	87.61	81.51	78.76	76.38	76.05	86.88	85.91	73.60	72.05
SSC ('21)	Resnet-50	81.87	86.87	82.77	**80.43**	–	–	86.52	86.99	–	–
DAFL ('22)	Resnet-50	83.54	**88.90**	83.72	80.29	–	–	87.07	86.40	–	–
Label2Label ('22)	Resnet-50	82.24	87.08	–	–	73.84	77.75	–	–	72.13	71.74
DVMIM (Ours)	Resnet-50	83.14	87.58	82.55	79.57	75.54	77.95	85.94	87.43	74.26	72.80
*DVMIM (Ours)	Resnet-50	83.60	87.74	82.95	79.70	75.70	**78.33**	86.11	**87.55**	74.75	**73.00**
PIL (Ours)	Resnet-50	85.63	86.98	84.90	78.12	79.73	75.20	87.82	85.72	75.25	71.84
*PIL (Ours)	Resnet-50	**86.32**	87.60	**85.45**	78.44	**80.29**	76.30	**88.45**	86.18	**76.19**	72.42
Baseline ('22)	ConvNeXt-base	82.2-	88.5-	–	–	76.54	**80.25**	86.1-	88.1-	75.21	75.62
ALM ('19)	BN-inception	80.68	86.46	81.87	80.16	74.28	76.65	84.24	85.41	73.01	71.53
MTA-Net ('20)	Resnet-152	–	–	77.62	79.07	–	–	84.62	86.04	–	–
AR-BiFPN ('21)	EfficientNet-B3	81.45	87.94	82.37	**82.33**	–	–	87.69	88.32	–	–
EALC ('23)	EfficientNet-B4	81.45	88.14	83.26	81.67	–	–	86.84	88.40	–	–
OAGCN ('23)	Resnet-101	84.43	86.80	86.02	75.90	76.20	76.03	88.21	87.32	75.44	73.35
DFDT ('23)	Swin Trans.	83.63	88.74	82.34	82.15	–	–	87.44	88.19	–	–
DVMIM (Ours)	ConvNeXt-base	86.55	89.42	84.54	81.08	78.60	79.86	87.48	**88.51**	78.09	75.76
PIL (Ours)	ConvNeXt-base	**88.70**	**89.53**	**86.35**	80.47	**82.04**	79.75	**89.09**	87.49	**79.73**	**75.81**

Highest metric values are given in bold

Our method DVMIM and PIL are compared with some notable approaches. For a fair comparison, we refer the baseline results on PA100k, RAP and PETA of Resnet-50 and ConvNeXt-base respectively from (Jia et al., 2021b) and (Specker et al., 2022), and the results of prior arts from (Jia et al., 2021b) or their original literature. For RAPzs and PETAzs, we adopt scores from the baseline (Jia et al., 2021b), or use the public code of previous work, if any, to reproduce the results. The baseline results of ConvNeXt-base on RAPzs and PETAzs are based on our experiments. If no result is reported as a certain setting or no public code is available for reproducing the results, it is marked as −. *results are produced with data augmentations strategy akin to JLAC. All values are percentages

5.2 Benchmark Results

We compare our methods with some notable state-of-the-art approaches, including MsVAA (Sarafianos & Kakadiaris, 2018), MTMS (Gao et al., 2019), ALM (Tang et al., 2019), MTA-Net (Ji et al., 2020), VAC (Guo et al., 2020), JLAC (Tan et al., 2020), AR-BiFPN (Moghaddam et al., 2021), SSC (Jia et al., 2021a), DAFL (Jia et al., 2022), Label2Label (Li et al., 2022), DFDT (Zheng et al., 2023), EALC (Weng et al., 2023) and OAGCN (Lu et al., 2023). Our training involves less data augmentation methods, which results in the baseline mean accuracy on all datasets of potentially up to 1% lower than some of the compared prior arts. To address this performance gap, we additionally present experimental results utilizing JLAC-style data augmentation settings (erasing, rotation, cropping, translation, additional random scaling and adding random Gaussian blurs). Table 1 reports the overall results, demonstrating that our approach of DVMIM achieves performance at least on par with others across all settings, while PIL consistently excels other methods with significant margins on mA. This not only proves the effectiveness of the proposed methods, but also validates our point of view that current models suffer from the accuracy degradation raising from the implicitly or explicitly modeled attributes co-occurrences. Notably, both PIL and DVMIM are implemented with minimal parameters, as in test time, the additional module over the baseline is only $\mathcal{H}_\psi(\cdot)$ of a single FC layer, in stark contrast to prior approaches that pay a premium regarding computational cost. Consequentially, the model parameters being largely saved by our lightweight models can be collected to incorporate an advanced-but-larger backbone, such as ConvNeXt, to achieve further significant improvements.

Also, the results indicate that our proposals perform better on PA100k compared to RAP and PETA. This outcome is expectable. As noted in the (Jia et al., 2021b), approximately 31.5% and 57.7% of pedestrian identities in the test set of RAP and PETA overlap with those in their respective training set. As a result, memorizing pattern of the biased

Table 2 Comparison of MixUp and our methods, and the breakdown effect for each technical contribution in the improved PIL

Method			mA	Prec.	Recall	F1
Baseline			80.38	87.09	87.01	87.05
MixUp			79.73	88.28	85.22	86.75
DVMIM (Ours)			83.14	86.76	88.40	87.58
PIL (Ours)			mA	Prec.	Recall	F1
NDSI	ABR	CRP				
			83.88	84.66	89.04	86.85
✓			84.53	84.89	89.13	87.01
✓	✓		**85.63**	84.05	**90.19**	86.98
✓		✓	83.29	**86.58**	88.85	**87.72**

Highest metric values are given in bold
NDSI is short for the norm-direction separated interpolation, ABR for the attributes-balanced re-sampling and CRP for the causal relations preservation

attributes co-occurrence in training set can be a form of information leakage about the test set, potentially causing current approaches overestimated on PETA and RAP. To address this issue, we also conduct experiments on PETAzs and RAPzs datasets, on which no identities from training set are overlapped in test set. Hence, the results on PETAzs, RAPzs and PA100k are more convincingly reliable and thereby possess more practical implications, on which our methods outperform other approaches considerably. It is also noteworthy that existing methods exhibit only trivial improvements on these realistic datasets, suggesting that prior developments in PAR on PETA and RAP may be partly attributed to better modeling of the co-occurrence bias common between training and test sets. Overall, our approaches excel in both recognition performance and practical applicability for realistic PAR.

5.3 Ablation Studies

We assess the functionality and characteristics of our methods. To better discern between models, we adopt PA100k and ResNet-50 in the following ablations. The overall comparison is presented in Table 2.

5.3.1 DVMIM Versus PIL

In this work, we have introduced two information-theoretic approaches to generate the attributes-disentangled features. As in Tables 1 and 2, both proposals enable a much accurate PAR, however, with distinctive characteristics.

Performance. The results in Tables 1 and 2 highlight that PIL surpasses DVMIM on mA and Recall. However, for Precision, PIL does not score on par with DVMIM in a major fraction of the cases. Such a performance divergence between DVMIM and PIL is not surprising since the classifier of PIL is optimized over the randomly produced labels, which endogenously change the source labels distribution. Considering that most attributes in dataset are under-represented, the shifted training label distribution would lead to more labels within the half zone of positive to be produced. Therefore, the classifier trained with PIL leans to predict more positives, leading to a higher Recall and lower Precision.

Training. For DVMIM, it can be practically difficult for D_θ to discern between $(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)$ sampled from the joint and the marginals, as both distributions of $P(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)$ and $P(\boldsymbol{y}^{/s}) \times P(\boldsymbol{f}^s)$ are ceaselessly shifted with the update of feature extractor, over the whole training phase. For this, we empirically set a smaller update step size for the generator by applying the loss weight λ in Eq. 4 with a default setting of 0.1. Additionally, before each epoch optimizing of Eq. 4, we train additional steps solely on $\{\mathcal{D}_{\theta^s}(\cdot,\cdot)\}_{s=1}^{C}$ (other modules are fixed) to make the Donsker-Varadhan represented KL divergence in Eq. 3 better converged to its upper bound, and thereby to enable $-\mathcal{L}_{dis}$ a looser estimator of the underlined mutual information. Also, to avoid the gradient explosion caused by the exponentiation in Eq. 3, we apply a moving average strategy akin to (Belghazi et al., 2018) for the back propagation of Eq. 4. Albeit effective, such operations hinder DVMIM training efficiency. While for PIL, its training is simple and efficient as given by Eq. 8.

DVMIM or PIL? One virtue of DVMIM is that the value of $-\mathcal{L}_{dis}$ in Eq. 6 can quantitively measure the inferences interdependencies among attributes. Hence, DVMIM not only provides an effective prescription for the attributes co-occurrence bias, but also a measurement of it. For PIL, when endued with the proposed improvements, it can mitigate the co-occurrence bias and imbalanced attributes distribution simultaneously, scoring substantially better than the prior arts and DVMIM, with both minimal training and test cost. Also, since the discriminators in DVMIM can still discern between posteriors by memorizing the joint distribution of labels in dataset, there exists a risk for DVMIM to over-reduce the feature factors, i.e., simply generate non-informative dummy features to minimize the mutual information to 0 (Fig. 9). Therefore, when Precision is not highly emphasized, PIL can be viewed as an all-around improved DVMIM.

Limitations. In summary, we highlight the limitations of both DVMIM and PIL as follows:

DVMIM: (1) DVMIM carries the risk of generating non-informative dummy features, which can drive the mutual information down to zero, rendering them ineffective; (2) Its training efficiency is notably poor; (3) DVMIM fails to achieve state-of-the-art performance.

PIL: (1) PIL is optimized based on randomly generated labels, inherently altering the original source label distribution, which compromises its performance in close-set settings; (2) The classifier trained using PIL tends to pre-

Table 3 Upper: Group splits of attributes in PA100k for causal attributes relations preservation. Lower: Within each group, we collect the test samples on which PIL predicts no or multiple attributes (irrational) and only one attribute (rational), and feed them respectively into a trained baseline model to compute the within-group averaged accuracy

Group index	Attributes
#1	ShortSleeve, LongSleeve
#2	UpperStride, UpperLogo, UpperPlaid, UpperSplice
#3	LowerStripe, LowerPattern
#4	Trousers, Shorts, Skirt&Dress
#5	AgeOver60, Age18–60, AgeLess18
#6	Front, Side, Back
Ungrouped	HoldObjectsInFront, Backpack, ShoulderBag, HandBag
Attributes	Boots, LongCoat, Glasses, Hat, Female

	#1	#2	#3	#4	#5	#6	Ungrouped
Rational	69.55	78.92	77.47	84.90	90.97	93.17	–
Irrational	28.80	11.94	27.88	18.73	19.20	22.24	–

The results indicate that, on samples that PIL gives irrational predictions, the accuracy of corresponding attributes drops drastically for the baseline method, revealing that the irrational outputs made by attributes-disentangled learning can be applied as cues suggesting for precaution of error-prone predictions, simply with some post hoc processings

dict more positive instances, resulting in a higher Recall but at the cost of a lower Precision.

5.3.2 PIL Versus MixUp

Although PIL optimizes over interpolated features with their correspondingly interpolated labels, Table 2 demonstrates that sample interpolation is not responsible for the significant performance boost, as MixUp yields only negative mA improvement over the baseline. Moreover, our method of DVMIM, which does not involve use of the fused features, performs comparably with the vanilla PIL, shown in Table 2. It indicates that the success of PIL lies in using feature interpolation to facilitate a differentiated in-distribution transform of each attribute specific feature, which is applied in Eq. 8 to decouple the shared feature factors.

5.3.3 Further Improvements of PIL

Norm-Direction Separated Interpolation. We use the proposed norm-direction separated interpolation for the feature transform in Eq. 8, in effect fully exploiting variations of attributes-specific features. The results in Table 2 show that PIL in conjunction with this technique outperforms that with plain feature interpolation. It not only proves the availability of this approach, but also implies that NDSI can be applicable to other fields encountering unstable image quality.

Attributes-Balanced Re-sampling. We study effect of changing the sampling function in Eq. (8) from within-batch random sampling to attributes-balanced sampling. From Table 2, when the generator converges, by simply altering to sample attributes features of label 0 and 1 with even probabilities, PIL is fostered greatly on the metrics of mA and Recall, however, with noticeable decrease of Precision. The reason is akin to that for PIL, for the majority under-represented attributes, the attributes-balanced re-sampling trains classifier with more features of positive label, making it tend to predict more positives than those really in the test set, which leads to higher Recall and lower Precision.

Preserving Causal Attributes Interdependencies. Intuitively, an ideal PAR should infer on causal attributes relations to robustify information exchange and propagation among attributes. For this, revisiting prior work that precisely models all attributes correlations seems suboptimal since without attributes-disentangled learning, such techniques would have difficulty in discerning the causal interdependencies from the intricate attributes co-occurrence bias and dismissing use of the latter only.

However, for PIL, we can simply set identical interpolation weights for the causally bound attributes to retain their relations in inference. Towards this aim, we manually organize the annotated attributes into different groups. The attributes within each group are regarded causally related, and thus share a common interpolation weight in Eq. 8. An example of group splits for PA100k is shown in Table 3, and we also compare the experimental results in it. As can be seen, model trained with causal relations preservation scores better on Precision with almost 2% improvement, while decreasing the mA. We take a closer look at this phenomenon and conjecture that *preserving any attributes interdependencies (even the causal ones) would inevitably harm model accuracy.* Take the attributes of Age$\leq$18, 18$<$Age$<$60, and 60$\leq$Age in PA100k as example, when the mutual exclusivity among them is retained, we find that model would not give irrational predictions like a person is both under 18 and older than 60. However, every time model fails at predicting the

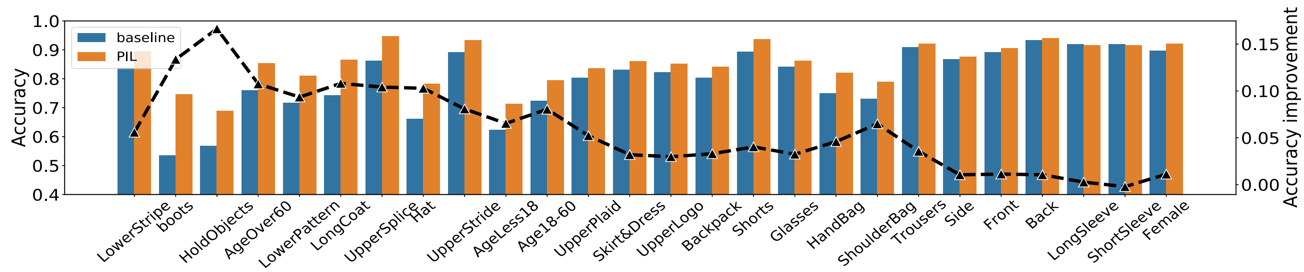

Fig. 8 Accuracy comparison between baseline and PIL on all attributes of PA100k. From left to right, attributes are sorted in a decreasing order of the distance between their respective positive ratio and 0.5. The black dashed line indicates the variation of accuracy improvement on each attribute, which is average smoothed with the windows size of 2 for better visualization

age, it makes mistakes at two age attributes simultaneously due to the mutual exclusivity between them. While for the vanilla PIL, if it encounters a hard sample for age prediction, to minimize the loss, it could just give no age predictions (or two predictions), making as few mistakes as possible by keeping the freedom to give irrational predictions. Therefore, conceptually, although the Precision can be facilitated by limiting model ability to give multiple positive predictions, the overall accuracy is impaired. Moreover, the irrational outputs are usually twinned with erroneous predictions, as shown in Table 3, taking advantage of which we can empower system to flag and abstain predictions from unknowns. As a conclusion, we argue that whether to apply causal relations preservation or not depends on the specific trade-off between sense-making and accurate intelligent system.

5.3.4 Experiments on Open-Set Datasets

We evaluated PIL on two cross-domain datasets to assess its generalization capabilities in scenarios where the interdependencies among attributes vary:

UPAR: In the context of open-set challenges in PAR, UPAR (Specker et al., 2022) consolidates the attribute annotations across four distinct PAR datasets. Specifically, models are trained on images sourced from a specific subset of these datasets and subsequently evaluated on an unseen dataset. Given that the public test set for UPAR has not yet been released, we have created a substitute dataset, termed UPAR*, based on its training data. To elaborate, we utilized the re-labeled PA100k, Market-1501, and PETA datasets from UPAR for training, while exclusively reserving the RAP dataset, which was left out, for testing purposes.

MSP60k: MSP60k (Jin et al., 2024) comprises 60,122 images and over 5000 unique person IDs, gathered through smart surveillance systems and mobile phones. These images encompass a wide range of domains and scenarios, including supermarkets, kitchens, construction sites, ski resorts, and various outdoor settings. The dataset is divided into two protocols: a random split and a cross-domain split. For our experiment, we have chosen to use the cross-domain setting.

The results are presented in Table 4, and it shows that the performance gap between PIL and prior arts is much larger for open-set testing, supporting the main idea of our work.

5.4 Discussion

Analysis on Improvements. Figure 8 presents the per-attribute accuracy of PIL on PA100. We observe sizable improvements on almost all attributes. It may relieve the likely concern that an indiscriminative discarding of attributes correlations could detriment the recognition of certain attributes. Notably, the over- or under-represented attributes (with positive ratio closer to 1 or 0) are more likely to be correlated with the occurrences pattern inherent to the insufficient images where they absent or present. On such attributes with most potential to be biased, our method facilitates the accuracy utterly by reducing the inference dependence of an imbalanced attribute to its frequently or hardly co-occurred attributes. While for attributes appearing evenly, the improvements are trivial.

Minimized Mutual Information. Performance gain of DVMIM and PIL demonstrates the efficacy of disentangling attributes. Here, with the mutual information estimator given in Eq. (3), we measure the amount of knowledge that is decoupled by our methods. Specifically, on the collected attributes-specific features produced by baseline (simply use a separate classifier on top of each attribute-specific feature to predict an attribute), DVMIM and PIL, we respectively train discriminators to discern between $P(\boldsymbol{y}^{/s}, \boldsymbol{f}^s)$ and $P(\boldsymbol{y}^{/s}) \times P(\boldsymbol{f}^s)$ by minimizing the $\mathcal{L}_{dis}$ stated in Eq. (6), and regard the converged upper bound of $-\mathcal{L}_{dis}$ as a loose neural estimate of $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s)$. In Fig. 9 we report the corresponding results. Compared to the baseline, both DVMIM and PIL significantly lower $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^s)$, showcasing that our methods successfully attenuate the common semantics among attributes-specific features. Interestingly, we also find that the baseline method produces attributes-specific features of

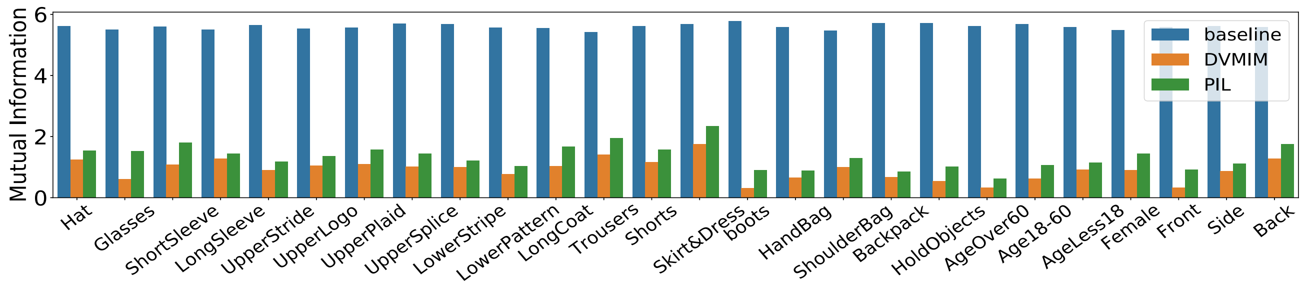

Fig. 9 For each attribute in PA100k, the estimated $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^{s})$ for the specific features produced by the baseline, DVMIM and PIL. It indicates that DVMIM and PIL significantly obviate the dependencies among attributes inferences

Table 4 Comparison of our method PIL against some notable approaches Label2Label (Li et al., 2022), OAGCN (Lu et al., 2023) and PARFormer-B (Fan et al., 2023) on the open-set datasets of UPAR* and MSP60k

Method		Baseline	Label2Label	OAGCN	PARFormer	PIL (Ours)
UPAR*	mA	69.76	70.44	73.94	72.58	**78.88**
	F1	78.60	78.25	79.53	78.86	**82.10**
MSP60k	mA	55.91	56.38	59.75	57.96	**62.47**
	F1	61.64	61.19	64.11	65.82	**69.84**

Highest metric values are given in bold

We use the public code released in prior arts literature to reproduce the results on UPAR*, and the results on MSP60k are from (Jin et al., 2024)

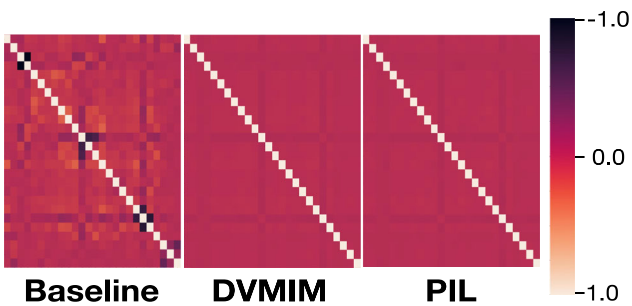

Fig. 10 The cosine similarity matrix of PA100k attributes anchor features for three compared methods. Classifier weights are adopted for attributes anchor features

almost identical $\mathcal{I}(\boldsymbol{y}^{/s}; \boldsymbol{f}^{s})$, revealing that they might be replication of each other in term of the encoded information, and thus not that distinctive and attribute-specific. Moreover, we also compute the cosine similarities between attributes anchor features of DVMIM, PIL and the baseline method in Fig. 10. The orthogonality between attributes features clearly supports that our methods eliminate the common feature factors among different attributes features.

Infestation of Attributes Independencies. Learning attributes co-occurrences has another drawback, for which we call the infestation of attributes independencies. As illustrated in Fig. 10, the pattern of interdependency of two attributes has an inclination of to be diffused by the model onto that of others. To be specific, two attributes strongly co-occur (or not) with each other would also lean to co-occur (or not) with other attributes, representing as the light or dark lines traversing the anchor features similarity matrix. Noticing that these spanning stripes do not appear in the Pearson correlation coefficient matrix of labels, but are by-produced during the modeling of attributes correlations. Our methods greatly, yet not totally, mitigate this tendency to hallucinate new interdependency bias from the dataset attributes co-occurrences, as there are still stripes can be eyed in the heat map.

6 Conclusions

To discard modeling of the attributes co-occurrence bias inherent to dataset, we have introduced a novel concept of attributes-disentangled feature learning that respects human cognition, which infers attributes without considering their interdependencies. Sequentially, it has been formulated as a problem of mutual information minimization and two information-theoretic regularizers, namely DVMIM and PIL, are proposed to enable that there are no feature semantics shared for recognizing different attributes. Also, we show that our method of PIL is highly extendable. With slight modifications on it, we can mitigate the accuracy degradation from imbalanced attributes distribution and preserve the causal attributes relations to further boost the reliability of PAR. Extensive experiments demonstrate our methods state-of-the-art performance and appealing merits, such as the ability to express ignorance and take precautions on uncertain predictions, better generalizability and practical applicability. Of paramount importance, our general perspective of attributes-disentangled learning differs from the paradigms of a plethora of previous methods, heuristically offering a promising avenue for future work of unbiased PAR.

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Data Availability For this paper only publicly available datasets were used.

Declarations

Code availability Public code for reproducing the benchmark results in original IJCAI paper is available at https://github.com/SDret/A-Solution-to-Co-occurence-Bias-in-Pedestrian-Attribute-Recognition

Conflicts of interest The authors have no conflicts of interest to declare that are relevant to the content of this article.

References

Belghazi, M. I., Baratin, A., Rajeswar, S., Ozair, S., Bengio, Y., Courville, A., & Hjelm, R. D. (2018). Mine: Mutual information neural estimation. arXiv preprint arXiv:1801.04062.

Chen, X., Duan, Y., Houthooft, R., Schulman, J., Sutskever, I., & Abbeel, P. (2016). Infogan: Interpretable representation learning by information maximizing generative adversarial nets. *Advances in Neural Information Processing Systems*, *29*.

Chen, Z., Li, A., & Wang, Y. (2019). A temporal attentive approach for video-based pedestrian attribute recognition. In *Pattern recognition and computer vision: second Chinese conference*, PRCV 2019, Xi'an, China, November 8–11, 2019, Proceedings, Part II 2, Springer, pp. 209–220.

Deng, Y., Luo, P., Loy, C. C., & Tang, X. (2014). Pedestrian attribute recognition at far distance. In proceedings of the 22nd ACM international conference on multimedia, pp. 789–792.

Do, K., & Tran, T. (2019). Theory and evaluation metrics for learning disentangled representations. arXiv preprint arXiv:1908.09961.

Donsker, M. D., & Varadhan, S. R. S. (1975). Asymptotic evaluation of certain Markov process expectations for large time, I. *Communications on Pure and Applied Mathematics, 28*, 1.

Fabbri, M., Calderara, S., & Cucchiara, R. (2017). Generative adversarial models for people attribute recognition in surveillance. In *2017 14th IEEE international conference on advanced video and signal based surveillance (AVSS)*, IEEE, pp. 1–6.

Fan, H., Hu, H. M., Liu, S., Lu, W., & Pu, S. (2020). Correlation graph convolutional network for pedestrian attribute recognition. *IEEE Transactions on Multimedia, 99*, 1–1.

Fan, X., Zhang, Y., Lu, Y., & Wang, H. (2023). Parformer: Transformer-based multi-task network for pedestrian attribute recognition. *IEEE Transactions on Circuits and Systems for Video Technology, 34*, 411–423.

Gao, L., Huang, D., Guo, Y., & Wang, Y. (2019). Pedestrian attribute recognition via hierarchical multi-task learning and relationship attention. *Association for Computing Machinery MM, 19*, 1340–1348. https://doi.org/10.1145/3343031.3351003

Guo, H., Zheng, K., Fan, X., Yu, H., & Wang, S. (2020). Visual attention consistency under image transforms for multi-label image classification. In *2019 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778.

Higgins, I., Matthey, L., Pal, A., Burgess, C., Glorot, X., Botvinick, M., Mohamed, S., & Lerchner, A. (2017). Beta-vae: Learning basic visual concepts with a constrained variational framework. In *International conference on learning representations*.

Jeon, I., Lee, W., Pyeon, M., & Kim, G. (2021). Ib-gan: Disentangled representation learning with information bottleneck generative adversarial networks. In *Proceedings of the AAAI conference on artificial intelligence,* vol. 35, pp. 7926–7934.

Jia, J., Chen, X., & Huang, K. (2021a). Spatial and semantic consistency regularizations for pedestrian attribute recognition. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)*, pp. 962–971.

Jia, J., Huang, H., Chen, X., & Huang, K. (2021b). Rethinking of pedestrian attribute recognition: A reliable evaluation under zero-shot pedestrian identity setting. arXiv preprint arXiv:2107.03576.

Jia, J., Gao, N., He, F., Chen, X., & Huang, K. (2022). *Learning disentangled attribute representations for robust pedestrian attribute recognition* (pp. 1069–1077). AAAI Press.

Ji, Z., Hu, Z., He, E., Han, J., & Pang, Y. (2020). Pedestrian attribute recognition based on multiple time steps attention. *Pattern Recognition Letters, 138*(2), 170–176.

Jin, J., Wang, X., Zhu, Q., Wang, H., & Li, C. (2024). Pedestrian attribute recognition: A new benchmark dataset and a large language model augmented framework. arXiv preprint arXiv:2408.09720.

Kauffmann, J., Ruff, L., Montavon, G., & Müller, K. R. (2020). The clever hans effect in anomaly detection. arXiv preprint arXiv:2006.10609.

Kinney, J. B., & Atwal, G. S. (2014). Equitability, mutual information, and the maximal information coefficient. *Proceedings of the National Academy of Sciences, 111*(9), 3354–3359.

Lampert, C. H., Nickisch, H., & Harmeling, S. (2009). Learning to detect unseen object classes by between-class attribute transfer. In *IEEE computer society conference on computer vision and pattern recognition (CVPR)*.

Lample, G., Zeghidour, N., Usunier, N., Bordes, A., Denoyer, L., & Ranzato, M. (2017). Fader networks: Manipulating images by sliding attributes. *Advances in Neural Information Processing Systems*, *30*.

Li, W., Cao, Z., Feng, J., Zhou, J., & Lu, J. (2022). Label2label: A language modeling framework for multi-attribute learning. In X. I. I. Part (Ed.), *Computer vision-ECCV 2022: 17th European conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings* (pp. 562–579). Springer.

Li, D., Chen, X., Zhang, Z., & Huang, K. (2017). Learning deep context-aware features over body and latent parts for person re-identification. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 384–393.

Li, D., Zhang, Z., Chen, X., Ling, H., & Huang, K. (2016). A richly annotated dataset for pedestrian attribute recognition. arXiv preprint arXiv:1603.07054.

Liu, P., Liu, X., Yan, J., & Shao, J. (2018). Localization guided learning for pedestrian attribute recognition. arXiv preprint arXiv:1808.09102.

Liu, Z., Mao, H., Wu, C. Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11976–11986.

Liu, X., Zhao, H., Tian, M., Sheng, L., Shao, J., Yi, S., Yan, J., & Wang, X. (2017). Hydraplus-net: Attentive deep features for pedestrian analysis. In *Proceedings of the IEEE international conference on computer vision*, pp. 350–359.

Li, D., Zhang, Z., Chen, X., & Huang, K. (2019). A richly annotated pedestrian dataset for person retrieval in real surveillance scenarios. *IEEE Transactions on Image Processing, 28*(4), 1575–1590.

Li, D., Zhang, Z., Shan, C., & Wang, L. (2023). Incremental pedestrian attribute recognition via dual uncertainty-aware pseudo-labeling. *IEEE Transactions on Information Forensics and Security, 18*, 2622–2636.

Lu, W. Q., Hu, H. M., Yu, J., Zhou, Y., Wang, H., & Li, B. (2023). Orientation-aware pedestrian attribute recognition based on graph convolution network. *IEEE Transactions on Multimedia, 26*, 28–40.

Meng, Q., Zhao, S., Huang, Z., & Zhou, F. (2021). Magface: A universal representation for face recognition and quality assessment. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp. 14225–14234.

Moghaddam, M., Charmi, M., & Hassanpoor, H. (2021). Jointly human semantic parsing and attribute recognition with feature pyramid structure in efficientnets. *IET Image Processing, 15*(10), 2281–2291.

Nguyen, X. L., & Wainwright, M. J. (2010). Estimating divergence functionals and the likelihood ratio by convex risk minimization. *IEEE Transactions on Information Theory, 56*(11), 5847–5861.

Ran, Y., Weiss, I., Zheng, Q., & Davis, L. S. (2007). Pedestrian detection via periodic motion analysis. *International Journal of Computer Vision, 71*(2), 143–160.

Ruderman, A., Reid, M., García-García, D., & Petterson, J. (2012). Tighter variational representations of f-divergences via restriction to probability measures. arXiv preprint arXiv:1206.4664.

Sarafianos, N., & Kakadiaris, I. A. (2018). Deep imbalanced attribute classification using visual attention aggregation. In *ECCV*.

Shalev, G., Adi, Y., & Keshet, J. (2018). Out-of-distribution detection using multiple semantic label representations. In *Advances in neural information processing systems*, pp. 7375–7385.

Shen, J., Guo, T., Zuo, X., Fan, H., & Yang, W. (2024). Sspnet: Scale and spatial priors guided generalizable and interpretable pedestrian attribute recognition. *Pattern Recognition, 148*, 110194.

Specker, A., Cormier, M., & Beyerer, J. (2022). Upar: Unified pedestrian attribute recognition and person retrieval. ArXiv: abs/2209.02522.

Tan, Z., Yang, Y., Wan, J., Guo, G., & Li, S. Z. (2020). Relation-aware pedestrian attribute recognition with graph convolutional networks. In *Proceedings of the AAAI conference on artificial intelligence,* vol. 34, No. (7), pp. 12055–12062.

Tang, C., Sheng, L., Zhang, Z. X., & Hu, X. (2019). Improving pedestrian attribute recognition with weakly-supervised multi-scale attribute-specific localization. In *2019 IEEE/CVF international conference on computer vision (ICCV)*.

Wang, X., Jin, J., Li, C., Tang, J., Zhang, C., & Wang, W. (2024). Pedestrian attribute recognition via clip based prompt vision-language fusion. *IEEE Transactions on Circuits and Systems for Video Technology*.

Wang, J., Zhu, X., & Gong, S. (2016). Video semantic clustering with sparse and incomplete tags. In *Proceedings of the thirtieth AAAI conference on artificial intelligence (AAAI-16)*.

Wang, F., Cheng, J., Liu, W., & Liu, H. (2018). Additive margin softmax for face verification. *IEEE Signal Processing Letters, 25*, 926–930.

Wang, X., Zheng, S., Yang, R., Zheng, A., Chen, Z., Tang, J., & Luo, B. (2022). Pedestrian attribute recognition: A survey. *Pattern Recognition, 121*, 108220.

Wang, J., Zhu, X., & Gong, S. (2017). Discovering visual concept structure with sparse and incomplete tags. *Artificial Intelligence, 250*, 16–36.

Weng, D., Tan, Z., Fang, L., & Guo, G. (2023). Exploring attribute localization and correlation for pedestrian attribute recognition. *Neurocomputing, 531*, 140–150.

Yan, X., Yang, J., Sohn, K., & Lee, H. (2016). Attribute2image: Conditional image generation from visual attributes. In *Computer vision–ECCV 2016: 14th European conference*, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part IV 14, Springer, pp. 776–791.

Yang, Y., Tan, Z., Tiwari, P., Pandey, H. M., Wan, J., Lei, Z., & Li, S. Z. (2021). Cascaded split-and-aggregate learning with feature recombination for pedestrian attribute recognition. *International Journal of Computer Vision, 129*, 1–14.

Yongqin, X., Lampert, C. H., Bernt, S., & Zeynep, A. (2017). Zero-shot learning-a comprehensive evaluation of the good, the bad and the ugly. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 99*, 1–1.

Yue, Z., Wang, T., Sun, Q., Hua, X. S., & Zhang, H. (2021). Counterfactual zero-shot and open-set visual recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 15404–15414.

Yutian, L., Liang, Z., Zhedong, Z., Yu, W., & Zhilan, H. (2019). Improving person re-identification by attribute and identity learning-sciencedirect. *Pattern Recognition,95*(C), 151–161.

Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2017). Mixup: Beyond empirical risk minimization. arXiv preprint arXiv:1710.09412.

Zhang, S., Chen, D., Yang, J., & Schiele, B. (2021). Guided attention in CNNs for occluded pedestrian detection and re-identification. *International Journal of Computer Vision, 129*, 1875–1892.

Zheng, A., Wang, H., Wang, J., Huang, H., He, R., & Hussain, A. (2023). Diverse features discovery transformer for pedestrian attribute recognition. *Engineering Applications of Artificial Intelligence, 119*, 105708.

Zhou, Y. (2022). Rethinking reconstruction autoencoder-based out-of-distribution detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.

Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., & Torralba, A. (2016). Learning deep features for discriminative localization. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2921–2929.

Zhu, J., Jin, J., Yang, Z., Wu, X., & Wang, X. (2023). Learning clip guided visual-text fusion transformer for video-based pedestrian attribute recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 2626–2629.

Zhu, X., Xu, C., & Tao, D. (2021). Where and what? examining interpretable disentangled representations. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 5861–5870.

Zhu, J., Liao, S., Lei, Z., & Li, S. Z. (2017). Multi-label convolutional neural network based pedestrian attribute classification. *Image Vision Computing, 58*, 224–229.

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